# ATM Guts Round

Paly Math Club

2025-26

Instructions:

- No materials are allowed, except for blank scratch paper, pens, pencils, and erasers.
- In particular, no calculators, no electronics, and no cheating.
- You should discuss with your team but may not discuss with others. (Teams can have up to 4 members).
- You may ask board members for clarification on problems, but it is up to the board's discretion whether to explain.
- The time limit is **30 minutes**.
- There will be 15 questions, in sets of 3. The last set is estimation.
- You must submit a set before getting the next one. You cannot go back to an earlier set.
- The test is free-response. Make sure to simplify answers completely.
- The sets are *very roughly* ordered in difficulty.
- The guts round is out of 177 points. Each problem in set 1 is worth 6 points, each problem in set 2 is worth 8 points, each problem in set 3 is worth 11 points, each problem in set 4 is worth 14 points, and each problem in the estimation set is worth up to 20 points.
- Estimation will be scored via a scoring function.
- Non-estimation problems give full points for correct answers and no points for leaving the question blank or wrong answers.
- There will be live scoring on a leaderboard.

Team name: _____

Team members:

_____

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**SET 1 [6 points each]:** (all answers are integers)

1. What are the last two digits of $7^{203}$?

2. A sequence is defined by $a_1 = 2, a_{n+1} = a_n^2 - a_n + 1$. Find

$$\sum_{i=1}^{10} \left( \frac{1}{a_i} \right)$$

to the nearest integer.

3. Medians $BD$ and $CE$ of a triangle $ABC$ are perpendicular, $BD = 16$, and $CE = 24$. Find the area of triangle $ABC$.

**SET 2 [8 points each]:** (all answers are integers)

4. Let triangle $ABC$ have side-lengths $AB = 13$, $BC = 14$, and $CA = 15$. Let $D$ and $E$ be the feet of the altitudes from vertices $A$ and $B$, respectively. Let the length of $DE$ be $\frac{m}{n}$. Find $m+n$.

5. Find the number of three-digit integers that contain at least one 0 or 5. The leading digit of the three-digit integer cannot be zero.

6. Find the number of polynomials $P(x)$ with coefficients in $\{0, 1, 2, 3\}$ such that $P(2) = 20$.

**SET 3 [11 points each]:**

7. Let triangle $ABC$ have $\overline{AB} = 5$, $\overline{BC} = 6$, and $\overline{CA} = 7$. Let the incircle of triangle $ABC$ be tangent to $BC$ at $D$. Let the $A$-excircle be the circle lying outside the triangle $ABC$ tangent to the segment $\overline{BC}$ and tangent to lines $\overleftrightarrow{AB}$ and $\overleftrightarrow{AC}$. The $A$-excircle of $ABC$ is tangent to $\overline{BC}$ at $E$. Find the length of $\overline{DE}$.

8. Let $P(x)$ be a polynomial of degree 100 such that $P(k) = \frac{1}{k}$ for $k = 1, 2, 3, \ldots, 101$. Find $\frac{1}{P(102)}$.

9. Suppose $a$, $b$, and $c$ be real numbers such that: $a^2$ - $bc$ = 14 $b^2$ - $ca$ = 14 $c^2$ - $ab$ = -3 Compute $|a + b + c|$.

**SET 4 [14 points]:**

10. The digits 0, 1, 2, 3, 4, 5, 6, 7, 8, 9 are each colored red, blue, or green. Find the number of colorings for which for any integer n $\leq$ 2: (a) If n is prime, then at least one digit of n is not blue. (b) If n is composite, then at least one digit of n is not green.

11. You have six blocks in a row, labeled 1 through 6, each with weight 1. Call two blocks x $\leq$ y connected when, for all x $\leq$ z $\leq$ y, block z has not been removed. While there is still at least one block remaining, you choose a remaining block uniformly at random and remove it. The cost of this operation is the sum of the weights of the blocks that are connected to the block being removed, including itself. Compute the expected total cost of removing all the blocks.

12. On plane $\alpha$, there are two points $A$ and $C$ with a distance of 4 between them, and a circle with center $C$ and radius 2. Let $B$ be the point of tangency of a tangent line drawn from $A$ to this circle. Point $P$ is located on a line passing through $B$ that is perpendicular to plane $\alpha$ such that $\overline{BP} = 2$. What is the distance between point $C$ and line $AP$?

**ESTIMATION SET [20 points]:**

13. Estimate E, the number of times an 8-digit number appears in Pascal's triangle.

14. A year is said to be interesting if it is the product of 3, not necessarily distinct, primes (for example $22 \cdot 5$ is interesting, but $22 \cdot 3 \cdot 5$ is not). How many interesting years are there between 5000 and 10000, inclusive?

15. Estimate the number of integers in {1, 2, 3, . . , 108} that can be written in the form $x^2$ - $2025y^2$ for some integers x and y. Submit a positive integer.